

Name: _____

Class: _____

Date: _____

Book Worksheet 4

The Electronic Age



Read: 1940s–1960s: wartime computers, transistors, and the race to the moon

The History of the Computer · Rachel Ignatofsky · Pages _____ to _____



The Big Picture

World War II pushed inventors to build huge electronic computers. Then transistors and microchips made computers smaller, faster, and cheaper.

Part 1 • Influential People

Fill in the table as you read.

Who / What	What did they create or do?	Why did it matter?
Alan Turing		
The ENIAC programmers Kay McNulty, Betty Jennings, Betty Snyder, Marlyn Wescoff, Fran Bilas, Ruth Lichterman		
Grace Hopper		
Bardeen, Brattain & Shockley		
Jack Kilby & Robert Noyce		
Margaret Hamilton		

Part 2 • Fill It In

Use each word from the word bank once.

vacuum tubes

moth

transistor

integrated circuit

Apollo

Bletchley Park

- During World War II, codebreakers at _____ worked to crack Germany's Enigma code.
- ENIAC used about 18,000 _____ and filled a whole room.
- In 1947, a real _____ stuck in the Harvard Mark II was taped into a logbook as the first "bug."
- The _____ replaced vacuum tubes because it was smaller, faster, and more reliable.
- An _____ (microchip) puts many transistors onto one small chip.
- Margaret Hamilton's software helped land _____ 11 on the moon in 1969.

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Check Your Understanding

Part 3 · Timeline

Order these events from earliest (1) to latest (7). Write the number in the box.

- | | |
|---|--|
| ☐ A. The transistor is invented | ☐ B. Apollo 11 lands on the moon |
| ☐ C. Colossus helps break codes in WWII | ☐ D. UNIVAC predicts the election on TV |
| ☐ E. Turing describes the “Turing machine” | ☐ F. ENIAC is completed |
| ☐ G. The integrated circuit is invented | |

Part 4 · Check Your Understanding

1. Why did the transistor change computing?

- A. It let computers connect to the Internet
- B. It replaced the need for programs
- C. It was smaller, faster, and more reliable than vacuum tubes, so computers could shrink
- D. It made computers run on steam

2. What does a compiler (like Grace Hopper's) do?

- A. Removes moths from computers
- B. Stores files when the power is off
- C. Connects computers into a network
- D. Translates code people can read into instructions the machine can run

3. Which is a computing innovation first built for war that later helped everyday life?

- A. The abacus
- B. Jacquard's loom
- C. Electronic computers built to break codes and calculate artillery tables
- D. The soroban

Part 5 · Stories from History

Pick one story or person from this section that surprised you. **Who** was involved, **what** happened, and **why** does it still matter?

Part 6 · Unit 5.3 Connection

The six women who programmed ENIAC were left out of photos and stories for decades. How is leaving people out of history similar to **computing bias**? What could happen if programmers today all come from the same background?
